Assignment 1: Business Goal

1. Business Context

European retail banks face intense competition from digital challengers and margin pressure from low interest rates. Term deposits are a critical, low-cost funding instrument — clients who commit savings provide the bank stable liquidity to fund loans and improve net interest margin. Retaining these depositors is equally important: churn to competitors forces costlier wholesale borrowing and erodes margins. To grow term deposits, the Portuguese bank runs outbound telemarketing campaigns. However, industry conversion rates are below 15%, making mass calling expensive and intrusive. A data-driven machine learning approach can solve this — analyzing client demographics, financial profiles, and prior campaign interactions to predict likely subscribers, allowing the bank to target calls precisely, reduce churn risk, and maximize campaign returns.

2. Key Business Goals

- Increase campaign conversion rate by ≥ 20% relative to the prior campaign baseline, using the model's ranked contact list to replace broad outreach — increasing efficiency and maximizing ROI per campaign.
- Reduce total outbound calls by ≥ 25% while retaining ≥ 90% of subscriptions, lowering agent labor and campaign costs.
- Improve campaign ROI by ≥ 15% versus the prior campaign, maximizing profitability.

3. Problem Definition

The core problem is determining who to call: given what the bank knows about each client — their demographics, financial profile, and prior campaign interactions — can we identify, before placing a call, which clients are likely to subscribe to a term deposit (yes/no)? Today, the bank contacts clients broadly, resulting in low conversion, high cost, and poor ROI. The data science solution must build a model that predicts subscription likelihood for each client, enabling the campaign team to prioritize high-propensity contacts — directly driving the conversion rate goal, the call reduction goal, and the ROI improvement goal defined in Section 2.

4. Stakeholder Identification

The following stakeholders have a direct interest in the data science solution:

Stakeholder	Role	What They Care About	Business Goal
Campaign	Design and	Interpretability,	Receive a ranked contact list with

Stakeholder	Role	What They Care About	Business Goal
Managers	execute the telemarketing campaign	actionable output	score thresholds and feature importance to guide targeting decisions and messaging strategy
Telemarketing Agents	Make outbound calls to prospects	Productivity, higher close rates	Work only qualified leads — fewer wasted calls, higher individual conversion rate and better performance incentives
Compliance & Legal	Ensure regulatory and data-privacy adherence	GDPR fairness, auditability	Model must not use protected attributes discriminatorily; feature importance output must be fully auditable for regulatory review
Bank Customers	Recipients of telemarketing outreach	Relevance, privacy, respect	Only contacted when genuinely likely to benefit — reducing unwanted calls preserves relationship and reduces complaints

5. Success Metrics (KPIs)

The following KPIs are directly tied to the business goals above and define what success looks like from a business perspective.

KPI	Definition	Target	Tracked By
Conversion Rate	Subscriptions ÷ Contacts × 100	≥ 20% improvement vs. prior campaign	Post-campaign dashboard
Cost Per Acquisition (CPA)	Total campaign cost ÷ new accounts opened	≥ 25% reduction vs. prior campaign	Finance system
Lift at Top Decile	Subscribers in top-scored 10% vs. random baseline — aligned with Moro et al. 2014	≥ 3× lift	Data science team pre-launch
Campaign ROI	(Deposit revenue − campaign cost) ÷ campaign cost	≥ 15% improvement vs. prior campaign	Post-campaign dashboard

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